THYROID HORMONE

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THYROID GLAND

Thyroid is an endocrine gland situated at the root of the neck on either side of the trachea.

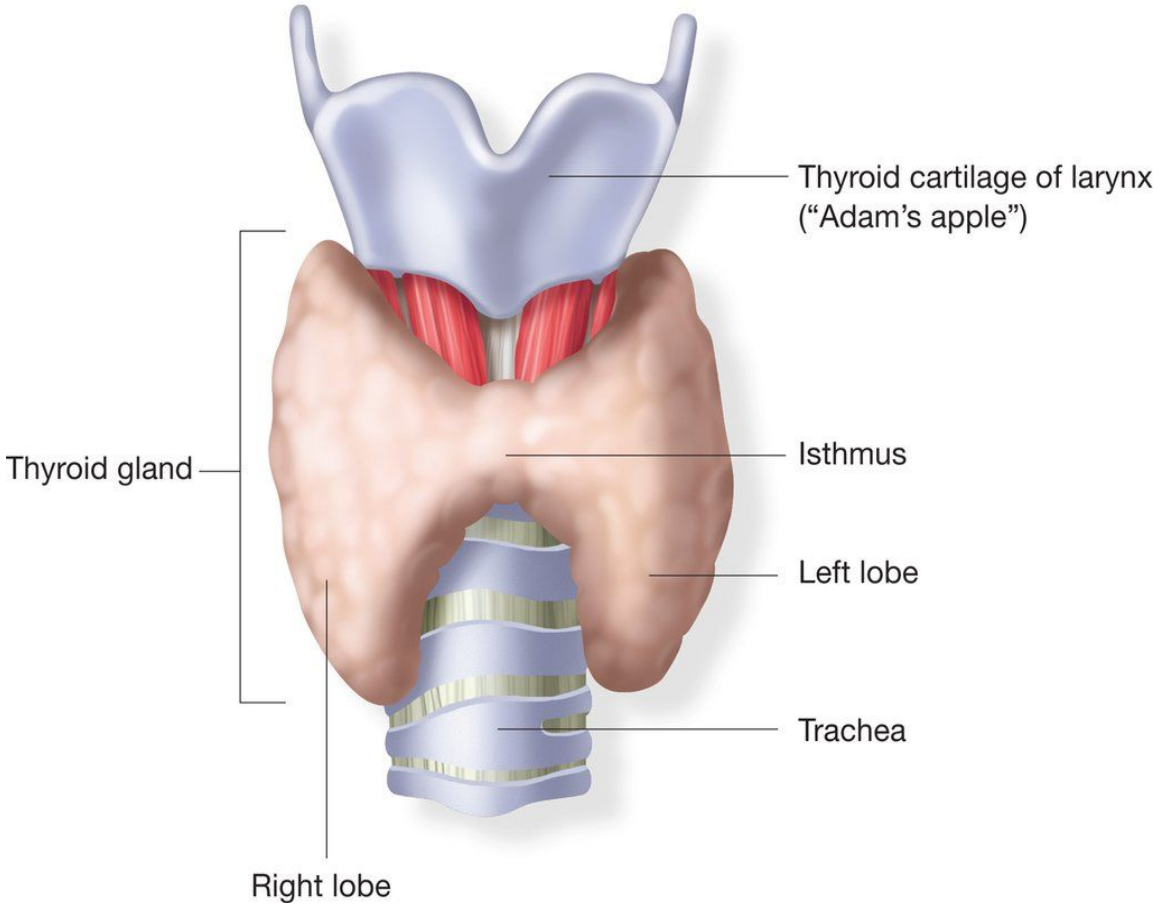
It has two lobes, which are connected in the middle by an isthmus.

It weighs about 15 to 20 g in adults.

Thyroid is larger in females than in males.

Thyroid hormones are synthesized and secreted by epithelial cells of the thyroid gland.

They have effects on virtually every organ system in the body including those involved in normal growth and development.



HISTOLOGY OF THYROID GLAND

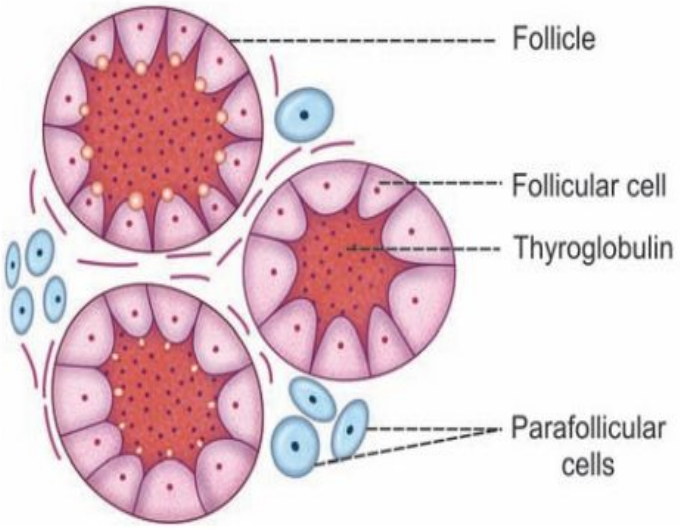


FIGURE 67.2: Histology of thyroid gland

- Thyroid gland is composed of large number of closed follicles. These follicles are lined with cuboidal epithelial cells, which are called the follicular cells.
- Follicular cavity is filled with a colloidal substance known as thyroglobulin, which is secreted by the follicular cells.
- Follicular cells also secrete tetraiodothyronine (T4 or thyroxine) and tri-iodothyronine (T3).
- In between the follicles, the parafollicular cells are present.
- These cells secrete calcitonin.

HORMONES OF THYROID GLAND

- Thyroid gland secretes three hormones:
  1. Tetraiodothyronine or T4 (thyroxine)
  2. Tri-iodothyronine or T3
  3. Calcitonin.
- T4 is otherwise known as thyroxine and it forms about 90% of the total secretion, whereas T3 is only 9% to 10%.

SYNTHESIS OF THYROID HORMONES

- Thyroid hormones are synthesized by the follicular epithelial cells of the thyroid gland. The follicular epithelial cells are arranged in circular follicles 200 to 300 μm in diameter.
- The material in the lumen of the follicles is colloid, which is composed of newly synthesized thyroid hormones attached to thyroglobulin. Synthesis of thyroid hormones takes place in thyroglobulin, present in follicular cavity.
- Iodine and tyrosine are essential for the formation of thyroid hormones. Iodine is consumed through diet. It is converted into iodide and absorbed from GI tract.

Tyrosine is also consumed through diet and is absorbed from the GI tract. For the synthesis of normal quantities of thyroid hormones, approximately 1 mg of iodine is required per week or about 50 mg per year.

STAGES OF SYNTHESIS OF THYROID HORMONES

- Synthesis of thyroid hormones occurs in five stages:

1. Thyroglobulin synthesis
2. Iodide trapping
3. Oxidation of iodide
4. Transport of iodine into follicular cavity
5. Iodination of tyrosine
6. Coupling reactions

1. Thyroglobulin Synthesis

- Thyroglobulin (TG), a glycoprotein containing large quantities of tyrosine, is synthesized on the rough endoplasmic reticulum and the Golgi apparatus of the thyroid follicular cells.
- Thyroglobulin is then incorporated into secretory vesicles and extruded across the apical membrane into the follicular lumen.
- After synthesis, thyroglobulin is stored in the follicle.
- Later, the tyrosine residues of thyroglobulin will be iodinated to form the precursors of thyroid hormones.

2. Iodide Trapping

- I^- is actively transported from blood into the follicular epithelial cells against both chemical and electrical gradientsThis process is called iodide trapping.
- Iodide is transported into the follicular cell along with sodium by sodium-iodide symport pump, which is also called iodide pump..
- The activity of this pump is regulated by I^- levels in the body. For example, low levels of I^- stimulate the pump. When there is a dietary deficiency of I^-, the Na^+ -I^- cotransport increases its activity, attempting to compensate for the deficiency.
- Normally, iodide is 30 times more concentrated in the thyroid gland than in the blood. However, during hyperactivity of the thyroid gland, the concentration of iodide increases 200 times more.

3. Oxidation of Iodide

- Iodide must be oxidized to elementary iodine, because only iodine is capable of combining with tyrosine to form thyroid hormones.
- The oxidation of iodide into iodine occurs inside the follicular cells in the presence of thyroid peroxidase.
- Absence or inactivity of this enzyme stops the synthesis of thyroid hormones.
- Thyroid peroxidase is inhibited by propylthiouracil (PTU), which blocks the synthesis of thyroid hormones by blocking all of the steps catalyzed by thyroid peroxidase. Thus, administration of PTU is an effective treatment for hyperthyroidism.

4. Transport of Iodine into Follicular Cavity

- From the follicular cells, iodine is transported into the follicular cavity by an iodide-chloride pump called pendrin.

5. Iodination of Tyrosine

- Combination of iodine with tyrosine is known as **iodination.** It takes place in thyroglobulin. First, iodine is transported from follicular cells into the follicular cavity, where it binds with thyroglobulin. This process is called **organification of thyroglobulin.**
- Iodination process is accelerated by the enzyme iodinase, which is secreted by follicular cells.
- At the apical membrane, just inside the lumen of the follicle, I2 combines with the tyrosine moieties of thyroglobulin, catalyzed by thyroid peroxidase, to form monoiodotyrosine (MIT) and diiodotyrosine (DIT). MIT and DIT remain attached to thyroglobulin in the follicular lumen until the thyroid gland is stimulated to secrete its hormones. High levels of I^- inhibit organification and synthesis of thyroid hormones, which is known as the Wolff-Chaikoff effect.

6. Coupling Reactions

- Iodotyrosine residues get coupled with one another. The coupling occurs in different configurations, to give rise to different thyroid hormones.
- **Coupling reactions** are:

i. One molecule of DIT and one molecule of MIT combine to form **tri-iodothyronine (T3)**

ii. Sometimes one molecule of MIT and one molecule of DIT combine to produce another form of T3 called **reverse T3** or rT3 (only 1% of thyroid output).

iii. Two molecules of DIT combine to form **tetraiodothyronine (T4)** which is thyroxine.

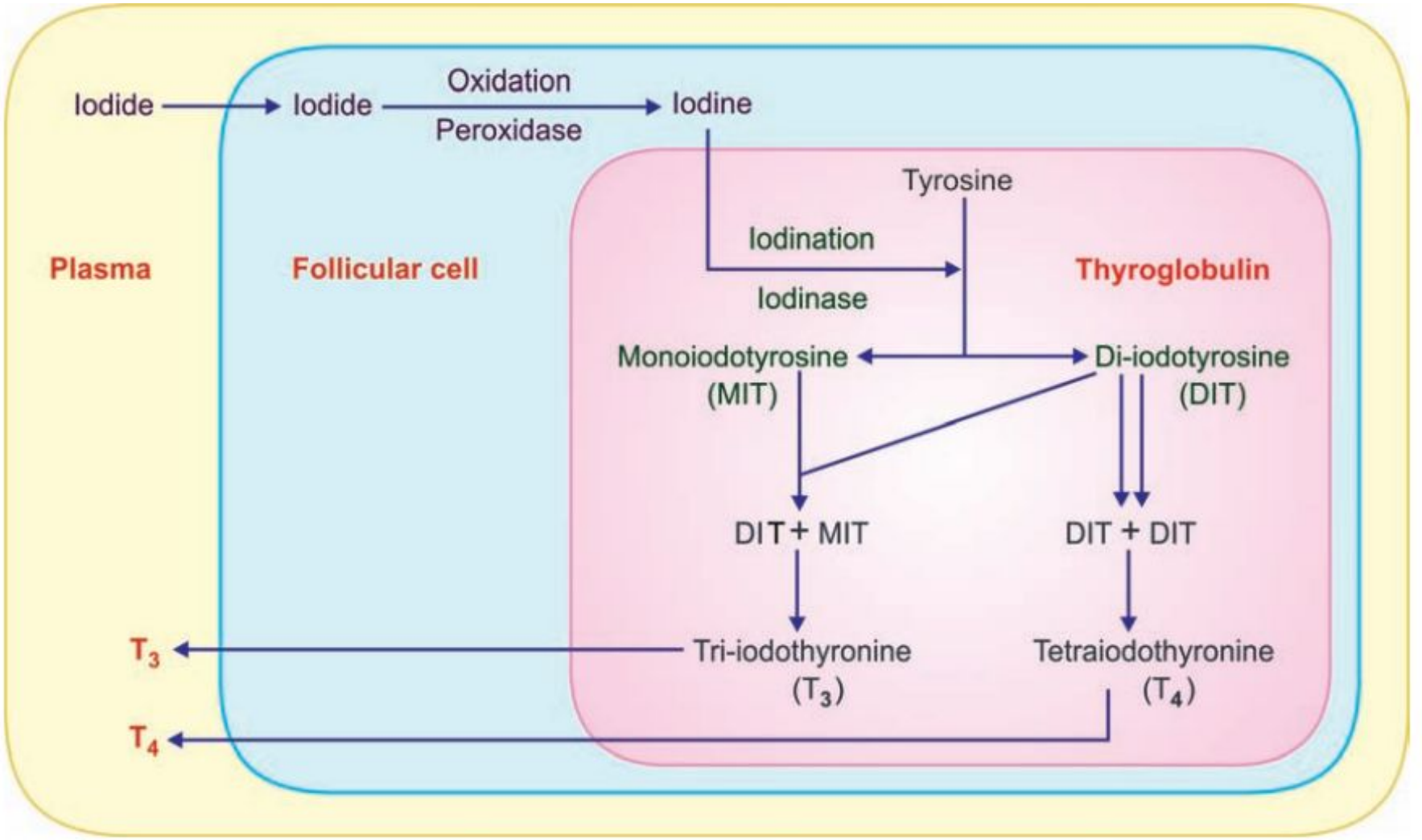
Tyrosine + I = Monoiodotyrosine (MIT)

MIT + I = Di-iodotyrosine (DIT)

DIT + MIT = Tri-iodothyronine (T3)

DIT + DIT = Tetraiodothyronine or Thyroxine (T4)

Synthesis of thyroid hormones



STORAGE OF THYROID HORMONES

- After synthesis, the thyroid hormones remain in the form of vesicles within thyroglobulin and are stored for long period. Each thyroglobulin molecule contains 5 or 6 molecules of thyroxine.
- In combination with thyroglobulin, the thyroid hormones can be stored for several months.
- Thyroid gland is unique in this, as it is the only endocrine gland that can store its hormones for a long period of about 4 months. So, when the synthesis of thyroid hormone stops, the signs and symptoms of deficiency do not appear for about 4 months.
- Thyroglobulin itself is not released into the bloodstream. On the other hand, the hormones are first cleaved from thyroglobulin and released into the blood.

FUNCTIONS OF THYROID HORMONES

- Thyroid hormones have two major effects on the body:

I. To increase basal metabolic rate

II. To stimulate growth in children.

Thyroid hormones act on virtually every organ system in the human body : Thyroid hormones act synergistically with growth hormone and somatomedins to promote bone formation; they increase basal metabolic rate (BMR), heat production, and oxygen consumption.

The actions of thyroid hormones are:

1. **ACTION ON BASAL METABOLIC RATE (BMR)**

Thyroxine increases the metabolic activities in most of the body tissues, except brain, retina, spleen, testes and lungs. It increases BMR by increasing the oxygen consumption of the tissues. The action that increases the BMR is called **calorigenic action.**

- In hyperthyroidism, BMR increases by about 60% to 100 above the normal level and in hypothythyroidism it falls by 20% to 40% below the normal level.

2. ACTION ON PROTEIN METABOLISM

- Thyroid hormone increases the synthesis of proteins in the cells. The protein synthesis is accelerated by the following ways:

i. By Increasing the Translation of RNA Thyroid hormone increases the translation of RNA in the cells. Because of this, the ribosomes are activated and more proteins are synthesized.

ii. By Increasing the Transcription of DNA to RNA Thyroid hormone also stimulates the transcription of DNA to RNA. This in turn accelerates the synthesis of proteins in the cells.

iii. By Increasing the Activity of Mitochondria In addition to acting at nucleus, thyroid hormone acts at mitochondrial level also. It increases the number and the activity of mitochondria in most of the cells of the body. It accelerates the synthesis of RNA and other substances from mitochondria, by activating series of enzymes. In turn, the mitochondria increase the production of ATP, which is utilized for the energy required for cellular activities.

3. ACTION ON CARBOHYDRATE METABOLISM

- Thyroxine stimulates almost all processes involved in the metabolism of carbohydrate.Thyroxine

i. Increases the absorption of glucose from GI tract

ii. Enhances the glucose uptake by the cells, by accelerating the transport of glucose through the cell membrane

iii. Increases the breakdown of glycogen into glucose

iv. Accelerates gluconeogenesis.

4. ACTION ON FAT METABOLISM

- Thyroxine decreases the fat storage by mobilizing it from adipose tissues and fat depots. The mobilized fat is converted into free fatty acid and transported by blood. Thus, thyroxine increases the free fatty acid level in blood.
- Even though there is an increase in the blood level of free fatty acids, thyroxine specifically decreases the cholesterol, phospholipids and triglyceride levels in plasma. So, in hyposecretion of thyroxine, the cholesterol level in plasma increases, resulting in **atherosclerosis**.
- Thyroxine also increases deposition of fats in the liver, leading to **fatty liver.**

6. ACTION ON BODY TEMPERATURE

- Thyroid hormone increases the heat production in the body, by accelerating various cellular metabolic processes and increasing BMR. It is called thyroid hormone induced thermogenesis.
- During hypersecretion of thyroxine, the body temperature increases greatly, resulting in excess sweating.

7. ACTION ON GROWTH

- Thyroid hormones have general and specific effects on growth. Increase in thyroxine secretion accelerates the growth of the body, especially in growing children.
- Lack of thyroxine arrests the growth. At the same time,thyroxine causes early closure of epiphysis. So, the height of the individual may be slightly less in hypothyroidism.
- Thyroxine is more important to promote growth and development of brain during fetal life and first few years of postnatal life. Deficiency of thyroid hormones during this period leads to **mental retardation.**
- Thyroxine is essential for maintaining the body weight. Increase in thyroxine secretion decreases the body weight and fat storage. Decrease in thyroxine secretion increases the body weight because of fat deposition.

8. ACTION ON CARDIOVASCULAR SYSTEM

- Thyroxine increases the overall activity of cardiovascular system.

i. On Heart Rate

Thyroxine acts directly on heart and increases the heart rate. It is an important clinical investigation for diagnosis of hypothyroidism and hyperthyroidism.

ii. On the Force of Contraction of the Heart

Due to its effect on enzymatic activity, thyroxine generally increases the force of contraction of the heart. But in hyperthyroidism or in thyrotoxicosis, the heart may become weak due to excess activity and protein catabolism. So, the patient may die of cardiac decompensation.

iii. On Arterial Blood Pressure

Because of increase in rate and force of contraction of the heart, increase in blood volume and blood flow by the influence of thyroxine, cardiac output increases. This in turn, increases the blood pressure.

9. ACTION ON GASTROINTESTINAL TRACT

- Generally, thyroxine increases the appetite and food intake. It also increases the secretions and movements of GI tract. So, hypersecretion of thyroxine causes diarrhea and the lack of thyroxine causes constipation.

10. ACTION ON CENTRAL NERVOUS SYSTEM

- Thyroxine is very essential for the development and maintenance of normal functioning of (CNS).

i. On Development of Central Nervous System

Thyroxine is very important to promote growth and development of the brain during fetal life and during the first few years of postnatal life. Thyroid deficiency in infants results in abnormal development of synapses, defective myelination and mental retardation.

ii. On the Normal Function of Central Nervous System Thyroxine is a stimulating factor for the CNS, particularly the brain. So, the normal functioning of the brain needs the presence of thyroxine.Thus, during the hypersecretion of thyroxine, there is excess stimulation of the CNS. So, the person is likely to have extreme nervousness and may develop psychoneurotic problems such as anxiety complexes, excess worries.

Hyposecretion of thyroxine leads to lethargy and somnolence (excess sleep).

11. ACTION ON SKELETAL MUSCLE

- Thyroxine is essential for the normal activity of skeletal muscles. Slight increase in thyroxine level makes the muscles to work with more vigor. But, hypersecretion of thyroxine causes weakness of the muscles due to catabolism of proteins. This condition is called **thyrotoxic myopathy.** The muscles relax very slowly after the contraction.
- Hyperthyroidism also causes fine muscular tremor. Tremor occurs at the frequency of 10 to 15 times per second. It is due to the thyroxine-induced excess neuronal activity, which controls the muscle. The lack of thyroxine makes the muscles more sluggish.

REGULATION OF SECRETION OF THYROID HORMONES

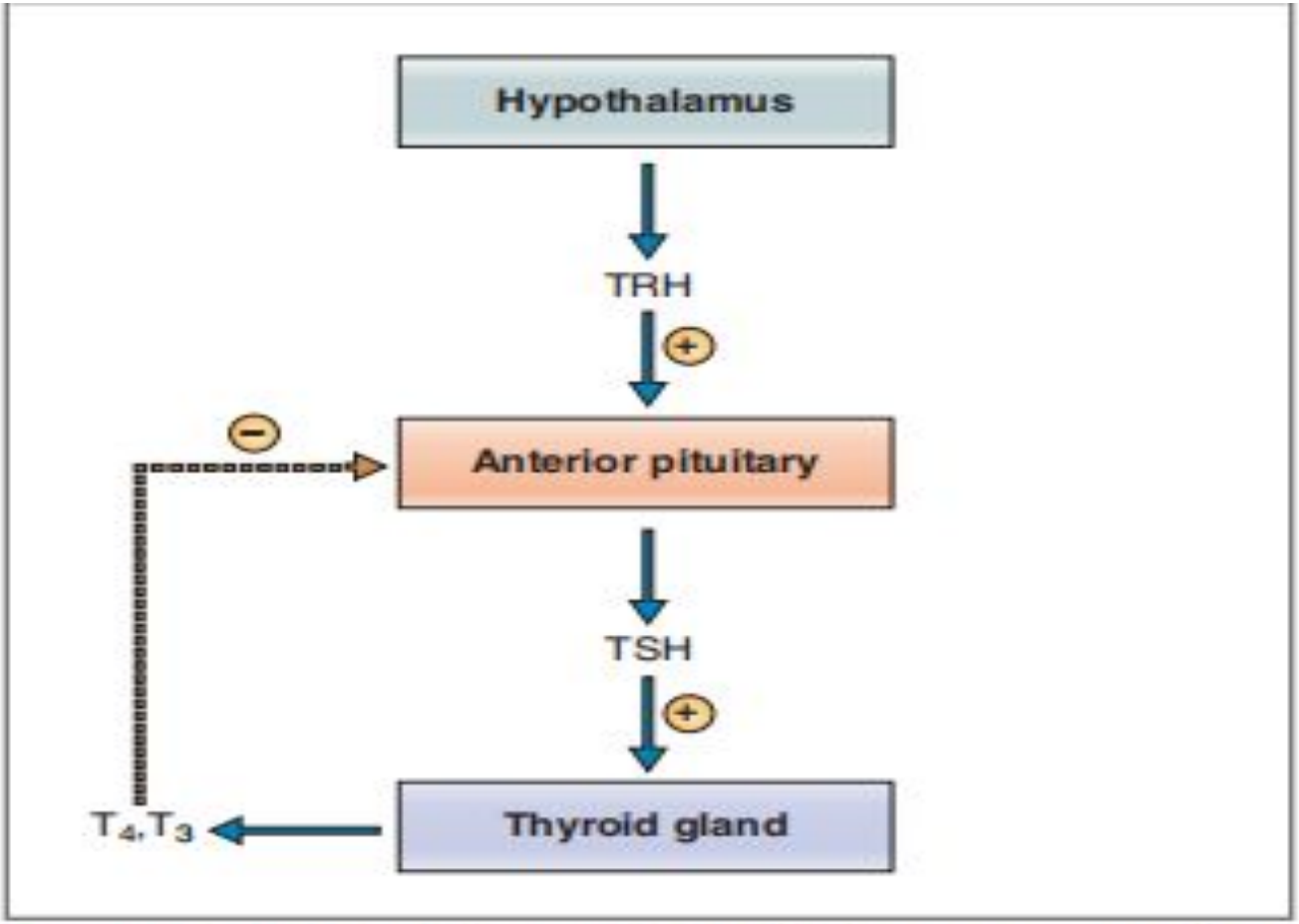
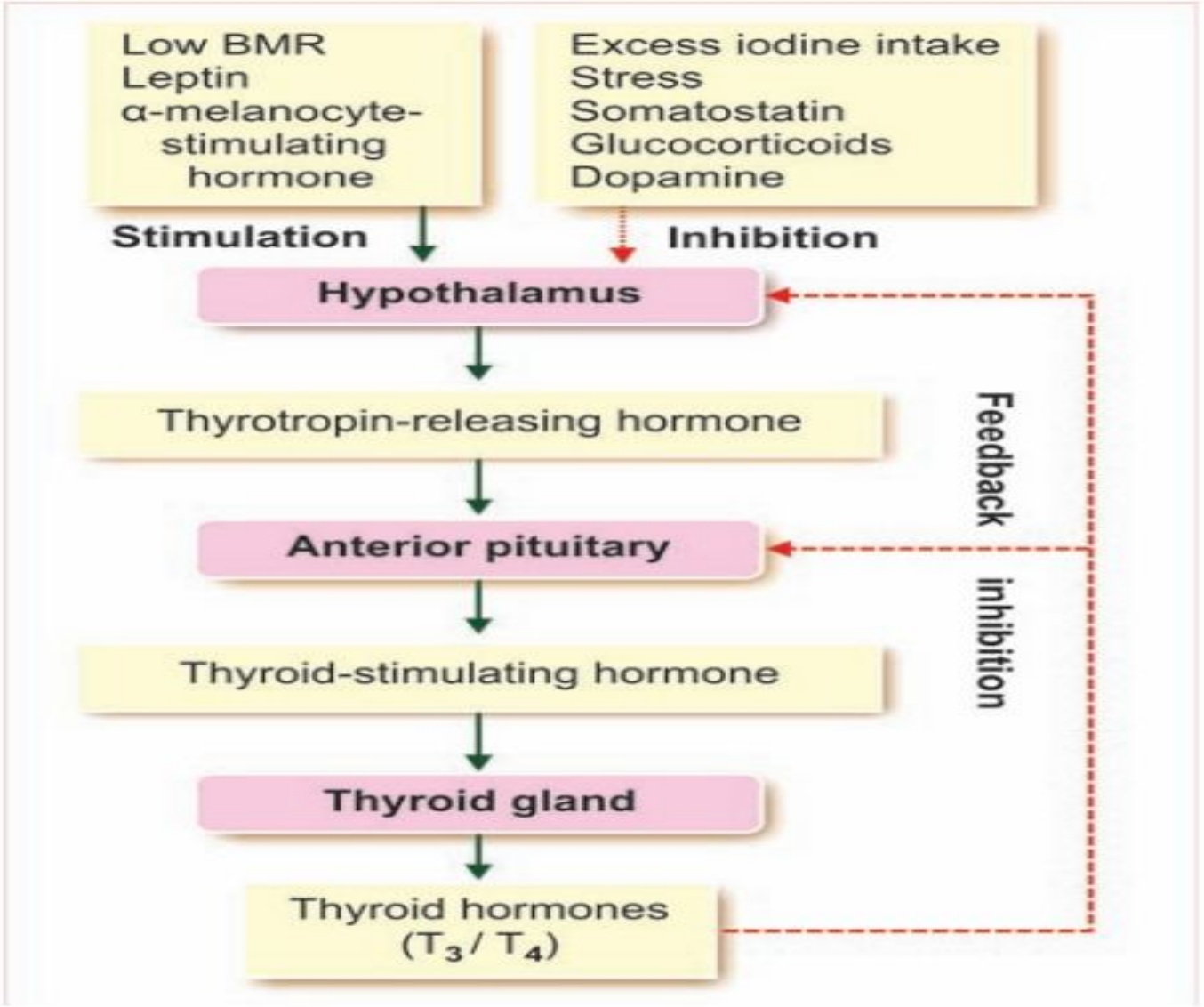


Figure 9–19 **Regulation of thyroid hormone secretion.** TRH, Thyrotropin-releasing hormone; TSH, thyroid-stimulating hormone; T_3, triiodothyronine; T_4, thyroxine.

REGULATION OF SECRETION OF THYROID HORMONES



REGULATION OF SECRETION OF THYROID HORMONES

- Secretion of thyroid hormones is controlled by anterior pituitary and hypothalamus through feedback mechanism. Many factors are involved in the regulation of thyroid secretion.

ROLE OF PITUITARY GLAND: Thyroid-stimulating Hormone

Thyroid-stimulating hormone (TSH) secreted by anterior pituitary is the major factor regulating the synthesis and release of thyroid hormones. It is also necessary for the growth and the secretory activity of the thyroid gland.

Mode of Action of TSH: TSH acts through cyclic AMP mechanism.

ROLE OF HYPOTHALAMUS

Hypothalamus regulates thyroid secretion by controlling TSH secretion through thyrotropic-releasing hormone (TRH). From hypothalamus, TRH is transported through the hypothalamo-hypophyseal portal vessels to the anterior pituitary. After reaching the pituitary gland, the TRH causes the release of TSH.

FEEDBACK CONTROL: Thyroid hormones regulate their own secretion through negative feedback control, by inhibiting the release of TRH from hypothalamus and TSH from anterior pituitary.

ROLE OF IODIDE: Iodide is an important factor regulating the synthesis of thyroid hormones. When the dietary level of iodine is moderate, the blood level of thyroid hormones is normal. However, when iodine intake is high, the enzymes necessary for synthesis of thyroid hormones are inhibited by iodide itself, resulting in suppression of hormone synthesis.


THANK YOU